WORKSHOP CANNY EDGE DETECTION

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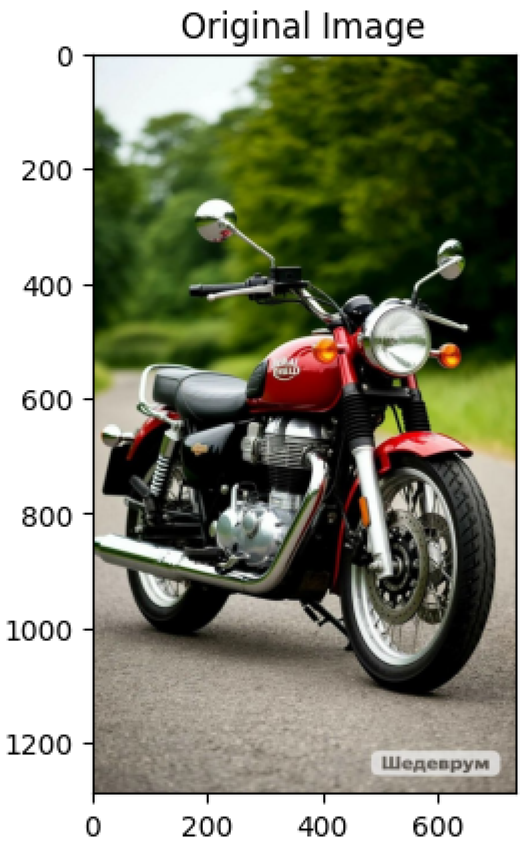
In [1]:
```python
import cv2
import matplotlib.pyplot as plt
```

In [2]:
```python
image = cv2.imread("pic bike.jpeg")
```

In [3]:
```python
plt.imshow(image[:,:,::-1])
plt.title("Original Image")
```

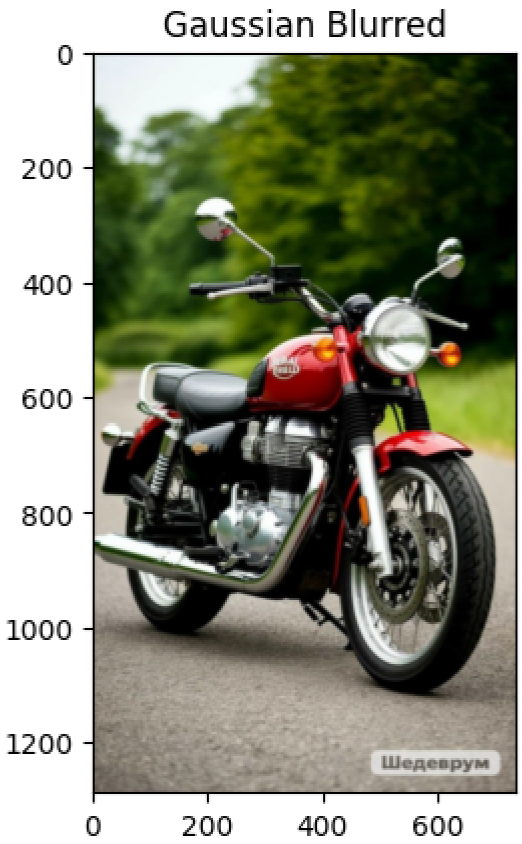
Out[3]:
```
Text(0.5, 1.0, 'Original Image')
```



In [4]:
```python
# Apply Gaussian Blur to reduce noise
blurred = cv2.GaussianBlur(image, (5, 5), 1.4)
```

In [5]:
```python
plt.imshow(blurred[:,:,::-1])
plt.title("Gaussian Blurred")
```

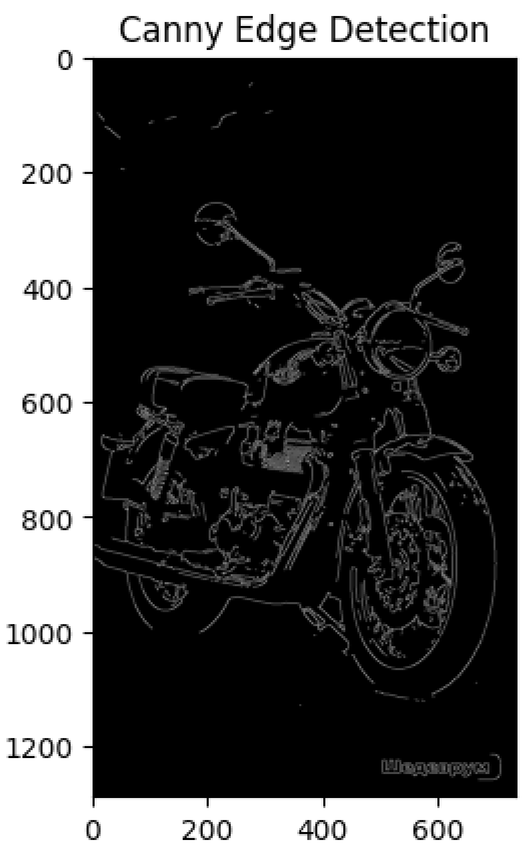
Out[5]:
```
Text(0.5, 1.0, 'Gaussian Blurred')
```



In [6]:
```
# Apply Canny Edge Detection
# Canny edge detection
edges = cv2.Canny(blurred, threshold1=180, threshold2=200)
```

In [7]:
```
plt.imshow(edges, cmap="gray")
plt.title("Canny Edge Detection")
```

Out[7]:
```
Text(0.5, 1.0, 'Canny Edge Detection')
```



In [8]:
```
# Comparison of Images
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(image[:,:,::-1])
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(blurred[:,:,::-1])
plt.title("Gaussian Blurred")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(edges, cmap="gray")
plt.title("Canny Edge Detection")
plt.axis("off")

plt.tight_layout()
plt.show()
```


```
In [ ]:
```